

ARTHUR AMALIR

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EDUCATION

Columbia University New York City, US
MS in Engineering Mechanics (Computational and Data-Driven Engineering), GPA: 3.95/4.0 Expected Dec 2026
Relevant Coursework: Numerical Analysis, Stochastic Engineering, Data Science & Machine Learning, Multiscale Computational Modeling, Predictive Modeling & Optimization, Random Signals & Noise

ESTP Cachan, FR
BS & MS in Engineering (Civil & Electrical), ECTS Grade A (Top 10%), Eq. GPA: 4.0/4.0 Expected Jun 2026
Relevant Coursework: Numerical Analysis, Probability & Statistics, Fluid and Solid Mechanics, Electrical Engineering and Energy Systems, Project Finance & Management, Structural Dynamics, Law

Lycée Janson de Sailly Paris, FR
MPSI & MP, GPA: 4.0/4.0 Jun 2022
Preparatory classes for national competitive entrance exams leading to French *Grande Ecoles*. Specialized in Math and Physics

EXPERIENCE

Columbia University | Department of Mechanical Engineering New York City, US
Research Intern - Uncertainty Quantification in Reduced Order Cardiovascular Models Jan 2026 - Apr 2026

- Extended deterministic cardiovascular simulations into a surrogate-based probabilistic risk assessment framework through uncertainty quantification and rare-event analysis.
- Achieved $>7,000\times$ computational savings over brute-force Monte Carlo while estimating $\sim 10^{-6}$ failure probabilities using Polynomial Chaos Expansion and Subset Simulation.

Columbia University | Department of Civil Engineering and Engineering Mechanics New York City, US
TA - Fluid Mechanics Sep 2025 - Dec 2025

- Evaluated assignments and led laboratory sessions, guiding students through experimental protocols and data analysis.

Columbia University New York City, US
Project - Multiscale Analysis of Diffusion in Random Networks Sep 2025 - Dec 2025

- Developed a scalable Monte Carlo PDE solver for diffusion on random 3D Voronoi networks, demonstrating that macroscopic transport is governed by connected 1D pathways despite 3D geometry.
- Developed a PyTorch Geometric GNN surrogate to predict diffusion speed from pore-throat graphs. Grade: 100.0/100.0

ESTP (with École des Ponts ParisTech & Agence de la transition écologique ADEME) Cachan, FR
Research Intern - IRGAK Project Jun 2025 - Aug 2025

- Cleaned, validated, and integrated experimental and simulation datasets for reliable parameter estimation
- Developed high-fidelity C++/FreeFEM++ multiphysics models calibrated against laboratory experiments and identified electrostatic interactions as the dominant transport-limiting mechanism in swelling clays, shaping future research objectives

ESTP Cachan, FR
Master's thesis - Tsunami Wave Propagation Modeling Sep 2024 - Jun 2025

- Optimized large-scale MATLAB numerical models simulating tsunami mitigation through sustainable natural barriers (coral reefs, mangroves), integrating rare-event and risk analysis to quantify extreme wave run-up behavior
- Captured reductions in run-up induced by sustainable natural barriers (40% lower run-up) with $100\times$ faster computation, validated against experimental datasets; evaluated by a jury of researchers and professionals. Grade: 17.0/20.0, highest in cohort

COLAS Cannes, FR
Intern Site Supervisor Jun 2023 - Aug 2023

- Supervised a team of 5-6 laborers on a public infrastructure project in Cannes
- Organized daily work plans and enhanced on-site efficiency. Acted as liaison between laborers, engineers, and municipal officials, strengthening project execution and maintaining timelines

SKILLS

Technical: PDE Modeling, Machine Learning, Monte Carlo Methods, Rare-Event Simulation, Surrogate Modeling, Numerical Optimization, Sensitivity Analysis
Software: Python (NumPy, SciPy, Pandas, Scikit-Learn, PyTorch), MATLAB, FreeFEM++/C++, ArcGIS, Microsoft Office Suite

EXTRA CURRICULAR ACTIVITIES

POLMUN ESTP (Politics & Model UN Society) Cachan, FR
Vice-President Sep 2024 - Jun 2025

- Organized and chaired university-level Model UN conferences and established partnerships with external universities in Paris

INTERESTS

Complex Systems, Technology and Society, Geopolitics, Napoleonic History, 19th-Century Literature, Philosophy, Chelsea FC